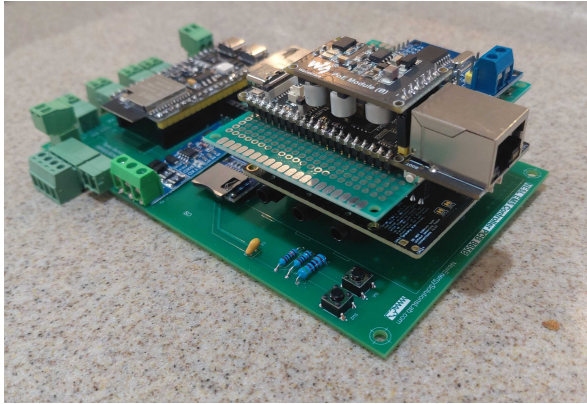


DTM-EMS

Transformer-edge intelligence for feeder-wide microgrid coordination, front and behind the meter

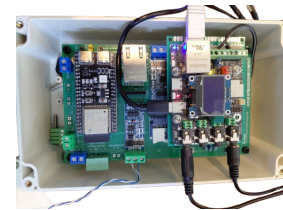


DTM-EMS is the intelligent controller at the heart of the secondary distribution transformer. It fuses transformer-grade polyphase precision metering ($\pm 0.1\%$) with neutral, phase and leakage insight, autonomous phase balancing and dynamic energy-policy enforcement, then **networks with multiple open-source Street-EMS nodes on a shared LV feeder** to coordinate the whole low-voltage network. Built on **open standards (IEEE, ISV, OpenAMI)**, it is the open, future-proof brain of the grid edge.

What Sets DTM-EMS Apart:

- ▶ **Precision metering + leakage insight** : $\pm 0.1\%$ polyphase, with neutral, phase and AC/DC leakage measured directly (plus NILM).
- ▶ **Autonomous phase balancing** : drives neutral current toward zero and cuts feeder losses, hands-off.
- ▶ **Feeder-wide coordination** : networks open-source Street-EMS nodes on one LV feeder for front-of-meter rural and urban village microgrids.
- ▶ **Front and behind the meter** : orchestrates behind-meter building EV charge / discharge and riser / balcony solar-and-battery.
- ▶ **Open-Source Hardware** : released under CERN-OHL-S v2 — schematics, board files and firmware published on GitHub (meshEMS / X-EMS).
- ▶ **Field-Serviceable** : built from open, off-the-shelf commodity modules, snap-in mounted for fast in-country assembly and repair.

One open, outdoor-rated box turns the distribution transformer into the coordinating brain of a feeder-wide microgrid, on standards you own.



Outdoor IP66 enclosure

Intelligent

- ▶ Neutral / phase / leakage + NILM insights
- ▶ Polyphase metering, $\pm 0.1\%$ active energy
- ▶ Autonomous phase balancing (neutral minimisation)

Interoperable

- ▶ Networks Street-EMS nodes on a shared LV feeder
- ▶ Open standards: IEEE / ISV / OpenAMI
- ▶ Ethernet / RS-485 / WiFi 6 / Thread / Zigbee / MQTT

Reliable

- ▶ Proven off-the-shelf metering hardware
- ▶ Onboard RTC, temperature & EEPROM
| stackable $\times 64$

Typical Applications

- ▶ Front-of-meter rural and urban village microgrid electrification
- ▶ Feeder-wide coordination of open-source Street-EMS nodes on a shared LV transformer
- ▶ Behind-the-meter building EV charge / discharge and riser / balcony solar-and-battery
- ▶ Neutral-current reduction, phase balancing and Type-B (AC+DC) leakage safety

PRODUCT	
Product name	DTM-EMS (10Power X-EMS, supplied by NESL)
Metering platform	Off-the-shelf polyphase precision-metering platform
Firmware	meshEMS-integrated open-source insights & policy layer
License & source	Hardware: CERN-OHL-S v2 Repository: GitHub, meshEMS / X-EMS
Serviceability	Open, off-the-shelf commodity modules snap-in mounted for fast in-country assembly and repair
METROLOGY	
Metering IC	Polyphase energy-metering IC
Channels	3 voltage inputs up to 4 CT: 3 phase + 1 neutral (clamp or Rogowski)
Accuracy	$\pm 0.1\%$ active energy $\pm 0.2\%$ reactive energy
High-current input	Rogowski-coil support for high-amperage feeders
INSIGHTS	
Insights engine	Neutral phase imbalance leakage NILM (non-intrusive load monitoring)
Measured	RMS V / I P / Q / S harmonics PF phase angle frequency bidirectional energy
POLICY ENFORCEMENT	
Edge enforcement	Dynamic energy-policy enforcement at the transformer edge
Phase balancing	Autonomous phase rebalancing to minimise neutral current and feeder losses
Feeder coordination	Networks with multiple open-source Street-EMS nodes on a shared LV feeder
INTEGRATION & NETWORKING	
Co-processor link	SPI + 40-pin expansion header (any MCU / Pi, stackable $\times 64$)
Networking	Ethernet (RJ45) RS-485 Modbus WiFi 6 / Thread / Zigbee / Matter
Telemetry	MQTT OpenAMI-compatible edge telemetry
Onboard	RTC temperature sensor EEPROM RGB status
POWER & MECHANICAL	
Power supply	5 V USB-C or 4.5-20 V DC or 8-12 V AC (SELV)
Dimensions	85 $\times$ 55 mm or Raspberry Pi HAT form factor
Enclosure	Outdoor IP66 subpanel enclosure converged EMS + DTM in one box
Mounting	Wall or DIN-rail outdoor mains subpanel
STANDARDS & POSITIONING	
Grid position	Secondary distribution transformer (DTM), coordinating the LV feeder
Open standards	IEEE ISV OpenAMI compatible
Deployment	Front-of-meter rural / urban village microgrids behind-meter EV charge / discharge riser / balcony solar + battery
Capabilities	Polyphase precision metering neutral / phase / leakage insight autonomous phase balancing feeder-wide coordination

Key Benefits

- ▶ Transformer-grade visibility: neutral, phase, AC/DC leakage and harmonics in one device
- ▶ Coordinates a whole LV feeder of open-source Street-EMS nodes, front and behind the meter
- ▶ Enables rural and urban village microgrid electrification and behind-meter EV, solar and battery
- ▶ Open standards (IEEE / ISV / OpenAMI): future-proof, with no vendor lock-in